

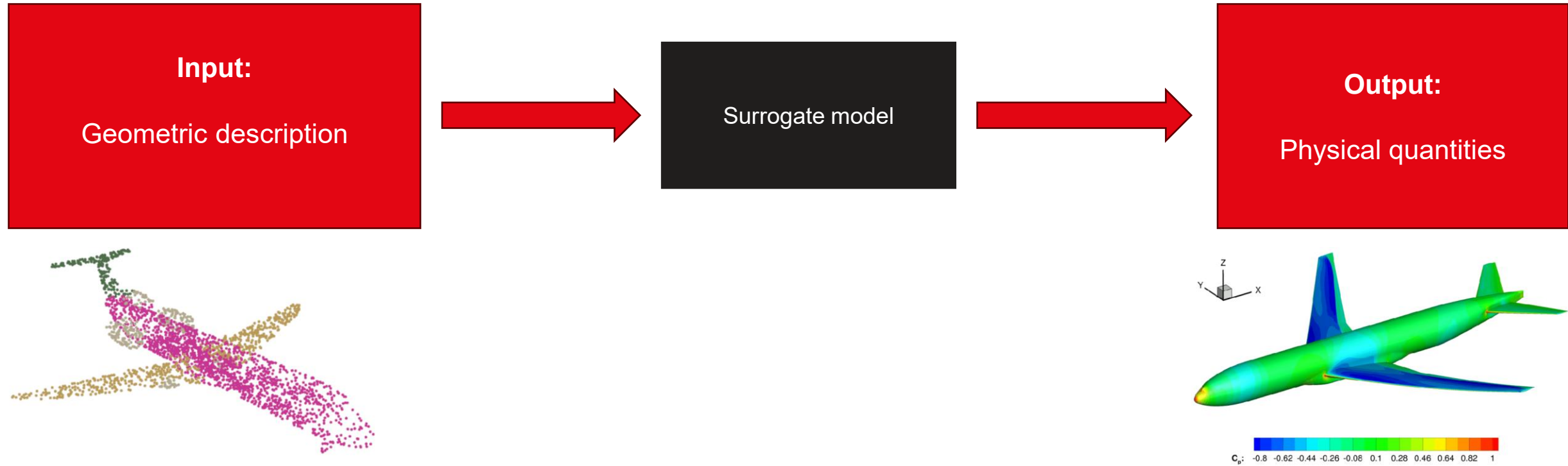
Pretrained 3D Foundation Models for Airfoil Surrogate Modeling

Semester Project at the CV Lab

Romain Corbel

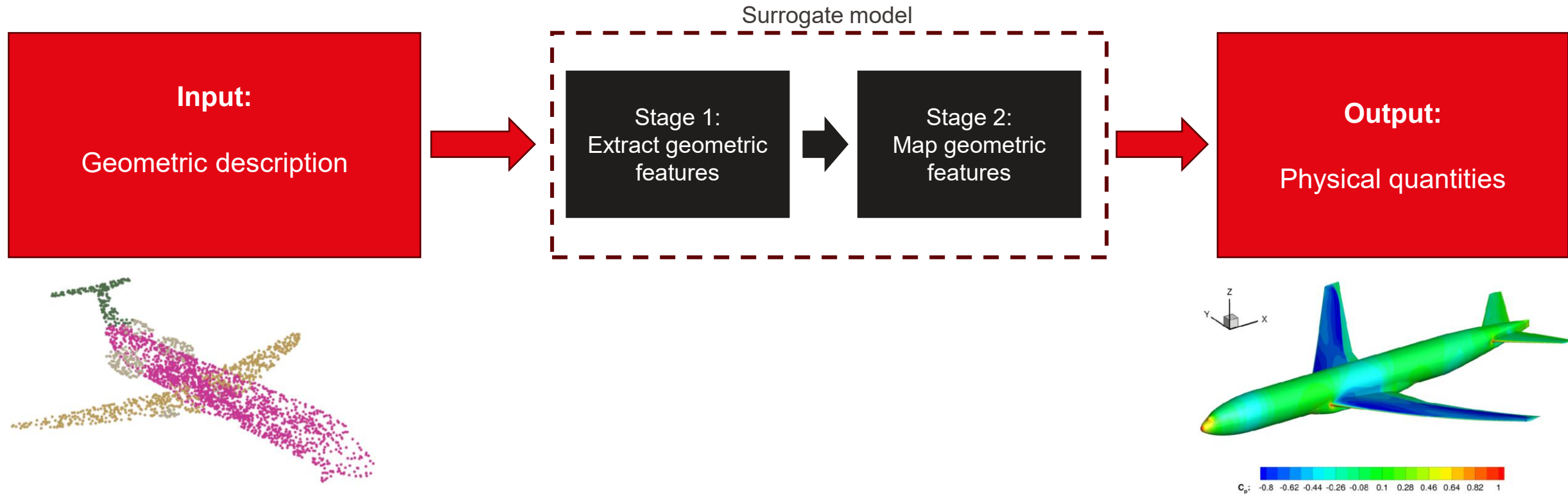
Supervised by Zhen Wei

CV Lab, Pascal Fua



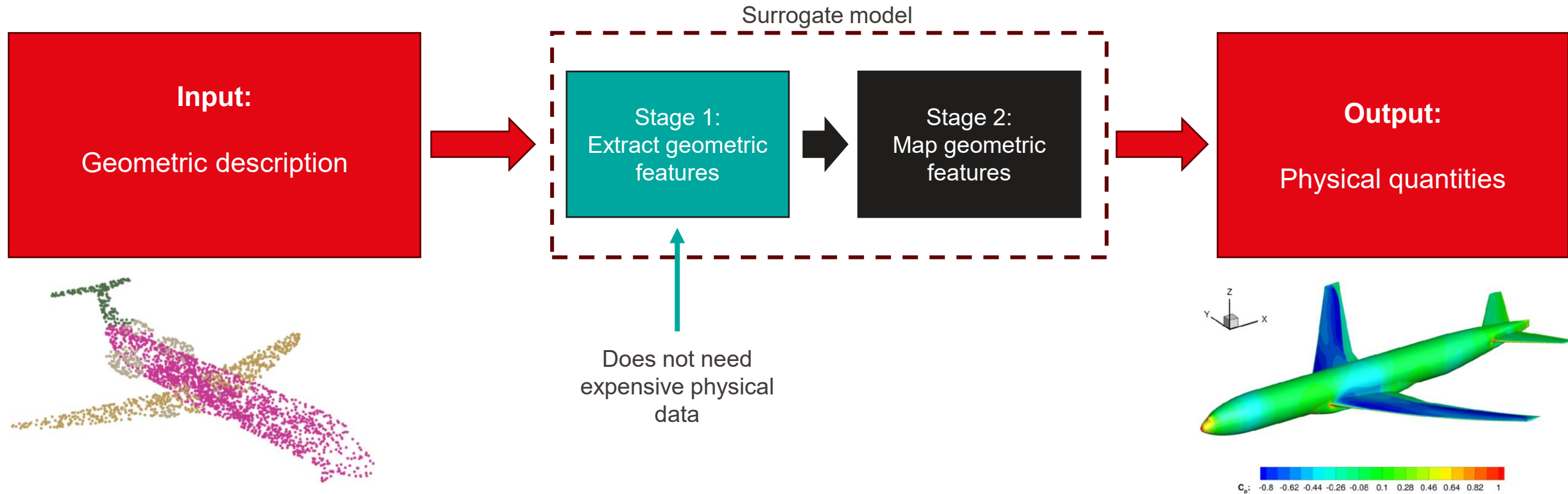
Situation:

- Aerodynamic design is governed by computationally intensive CFD solvers
- ➔ Surrogate models are needed for optimization
- ➔ But training data are expensive



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**Foundation models:**

“A *foundation model* is any model that is trained on broad data (generally using self-supervision at scale) that can be adapted (e.g., fine-tuned) to a wide range of downstream tasks”, Stanford Institute for Human-Centered Artificial Intelligence (HAI)

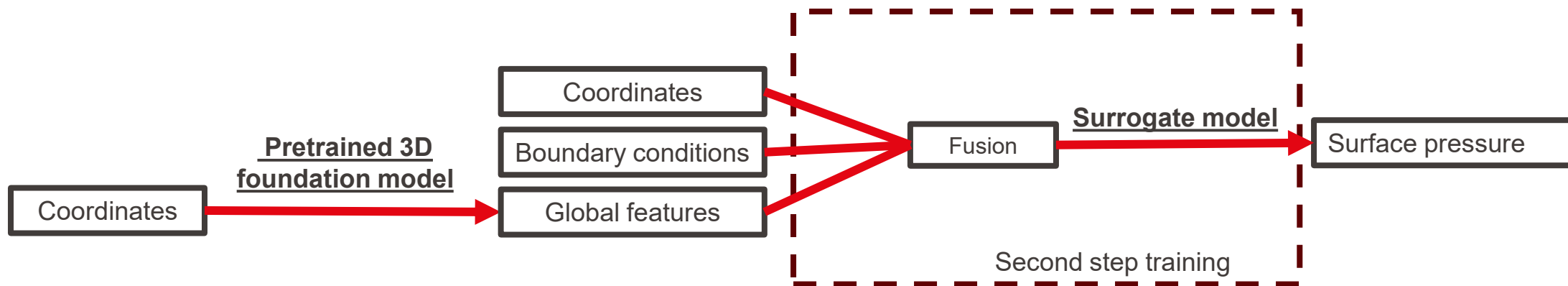
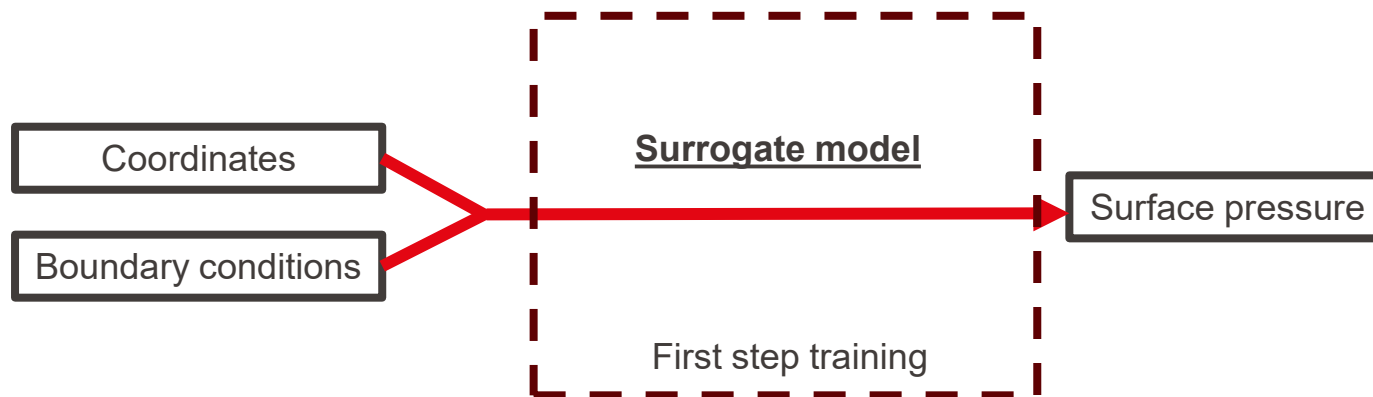
Idea: Decoupling Geometry Learning from Physics Data by using 3D Foundation Models

Research question: Can 3D Foundation Models improve surrogate models with limited CFD data?

General strategy:

1. Pipeline

Baseline pipeline:



- 4 surrogate model baselines: MLP, GraphSAGE, Graph U-Nets, Pointnet
- Trained on 200 airfoil's point clouds to predict surface pressure

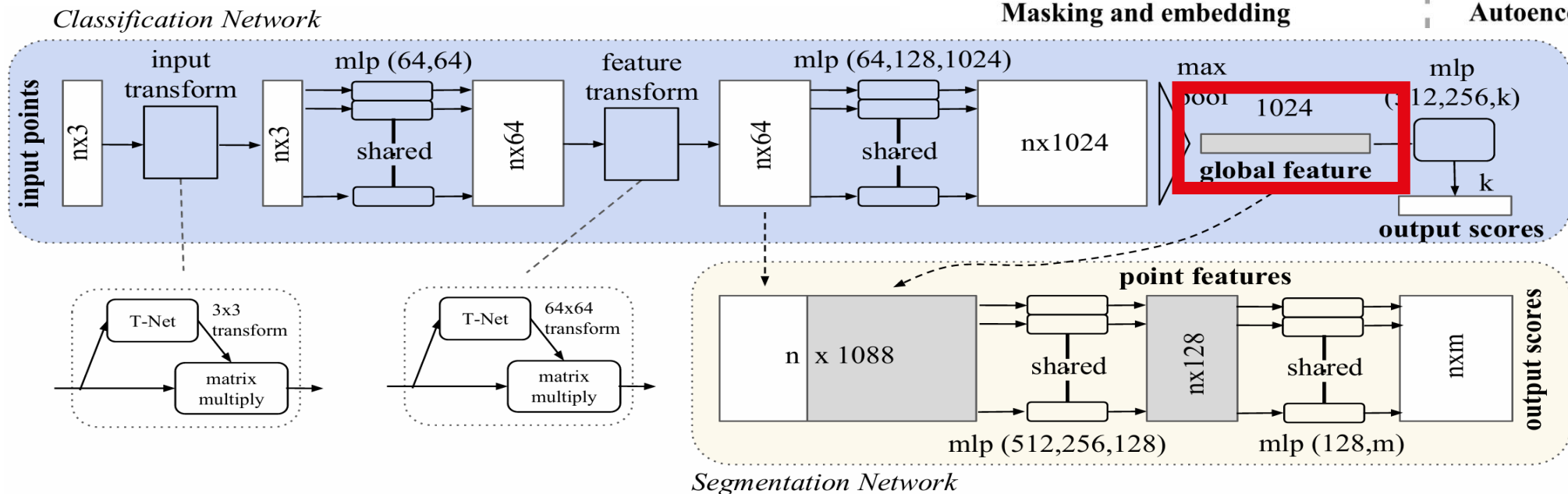


2. Feature extraction

ShapeNetPart Segmentation dataset

- 16 object categories, 2048 uniformly sampled points per point cloud

PointNet classification model:



I. Global feature fusion

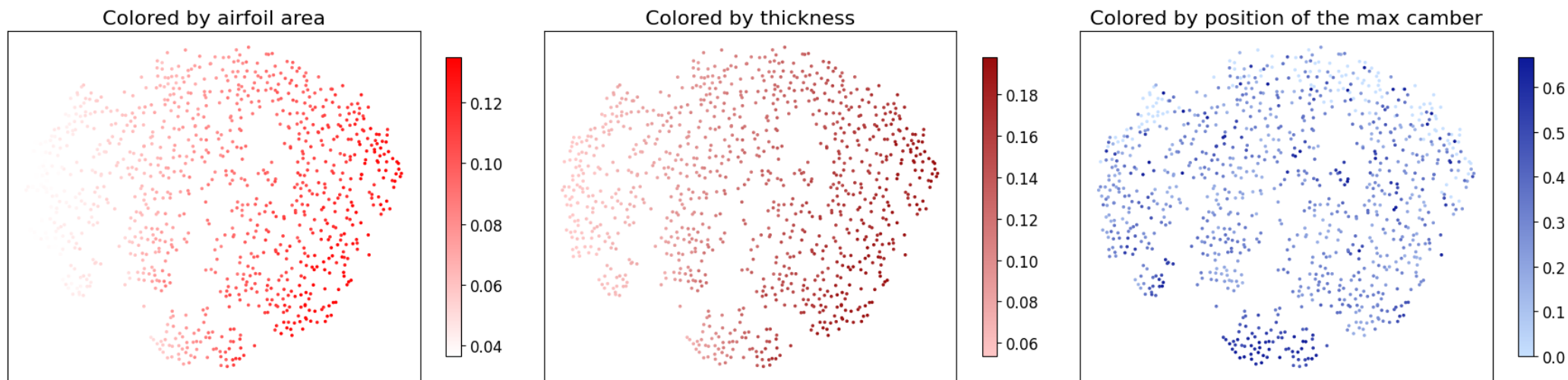
Methods:

- Visualization via t-SNE
- Color by area, thickness and position of the maximum camber

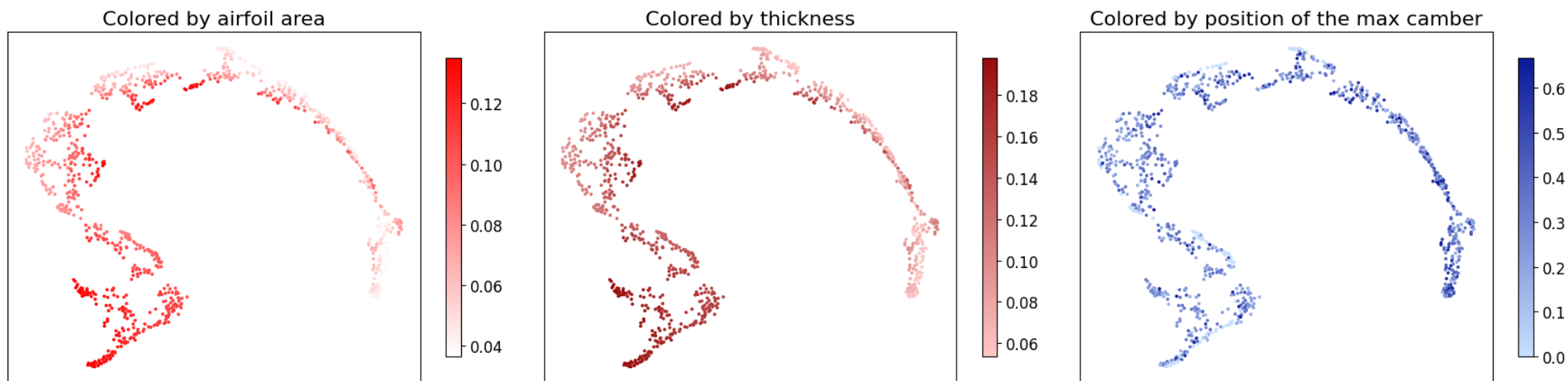
Expected results:

- Color gradients / clusters

t-SNE of extracted features from Point-MAE



t-SNE of extracted features from PointNet



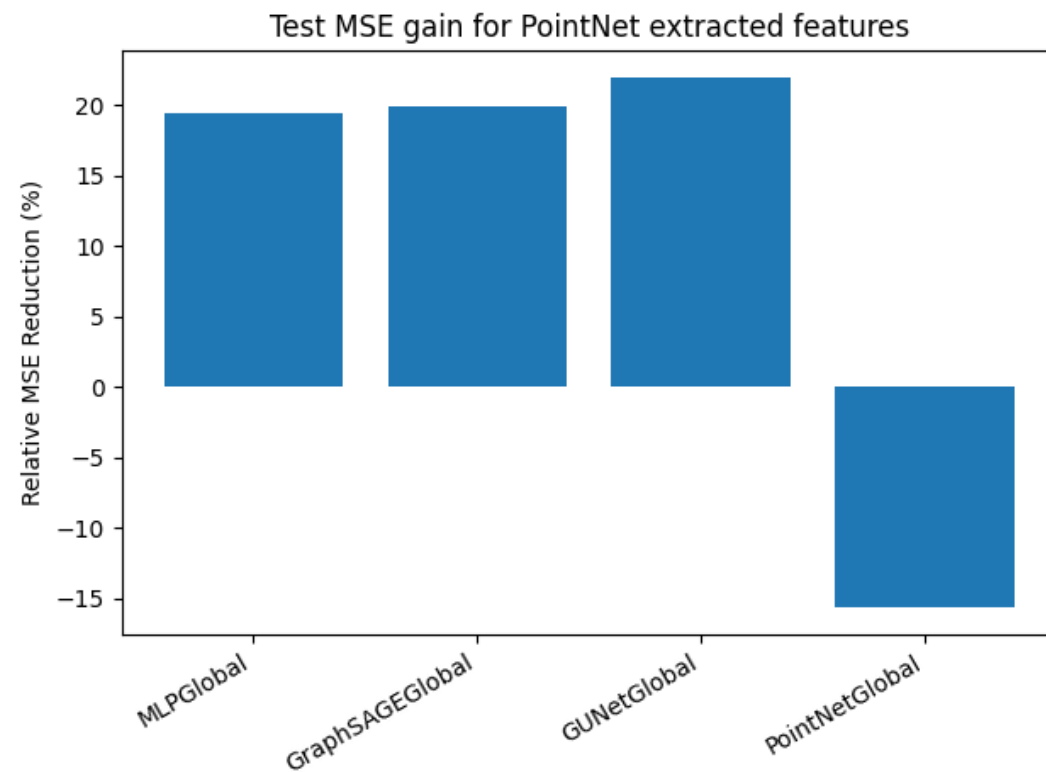
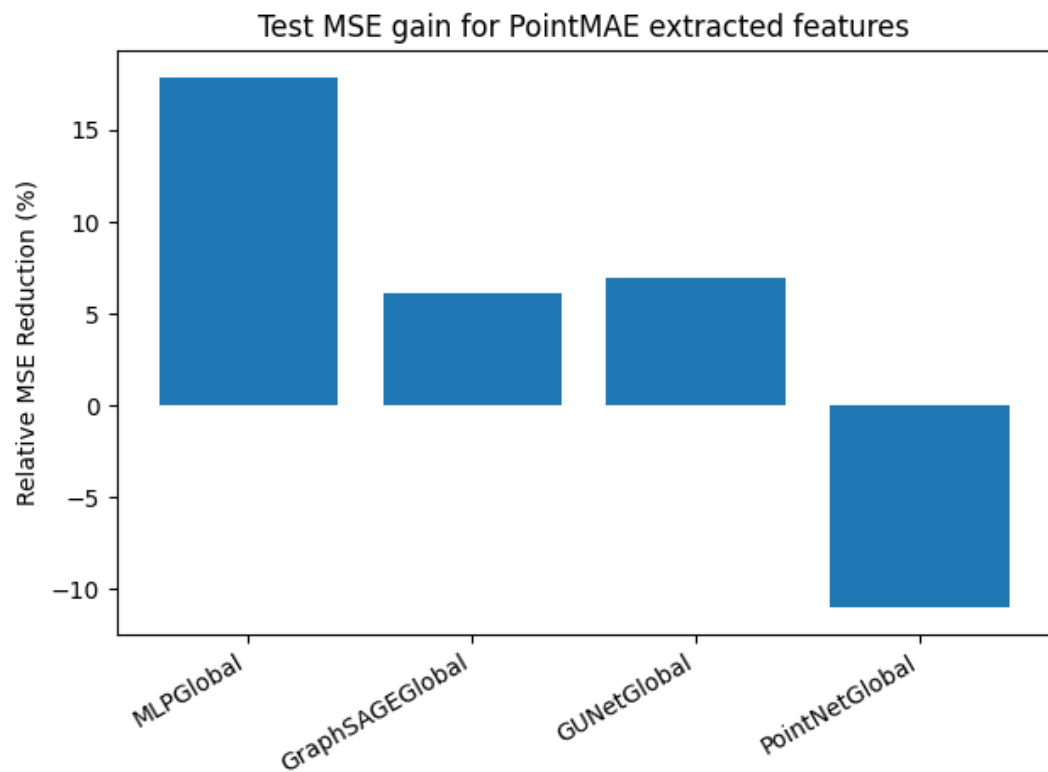
Method:

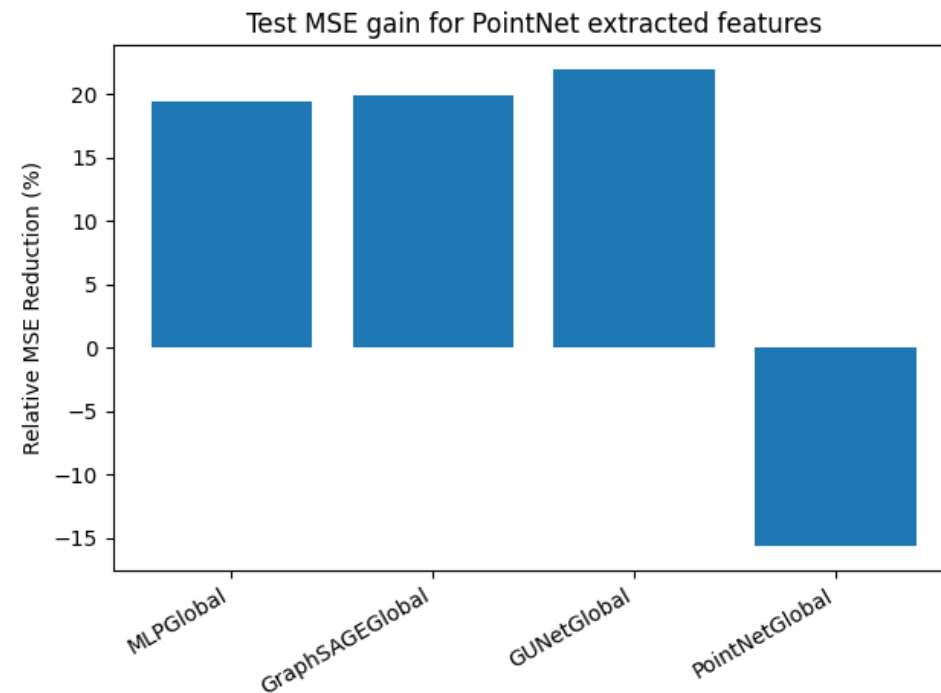
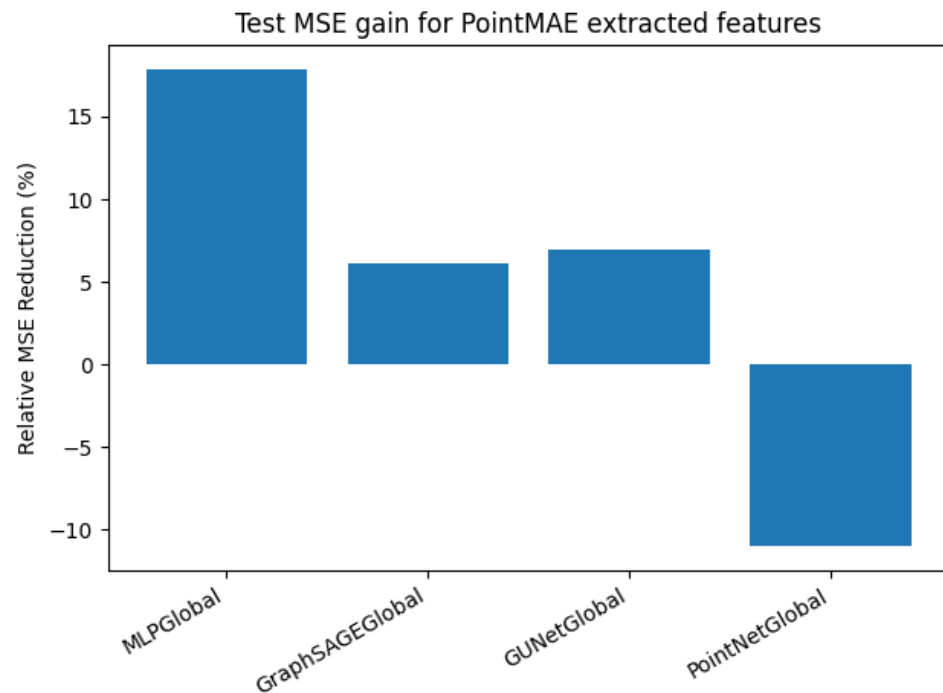
- Compare test loss gains over respective baselines

Expected results:

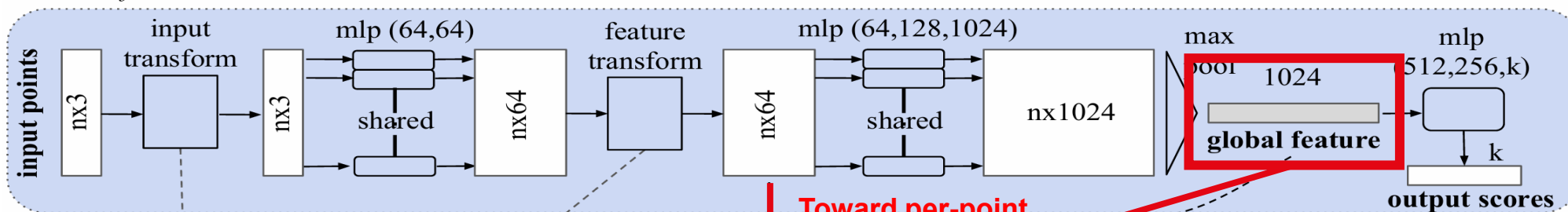
- Fusion is expected to reduce the test loss

$$\text{Relative MSE reduction (\%)} = \frac{\text{MSE}_{\text{baseline}} - \text{MSE}_{\text{global}}}{\text{MSE}_{\text{baseline}}} \times 100$$

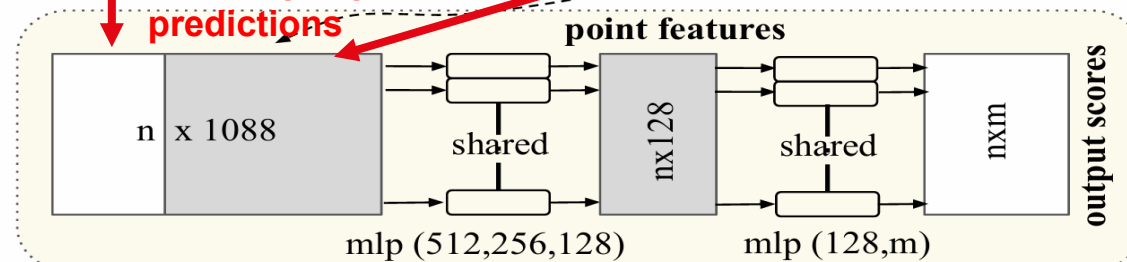
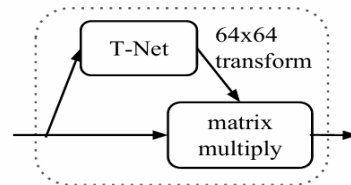
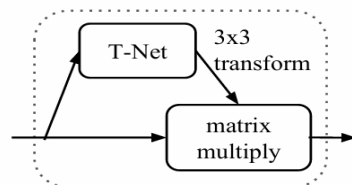




Classification Network

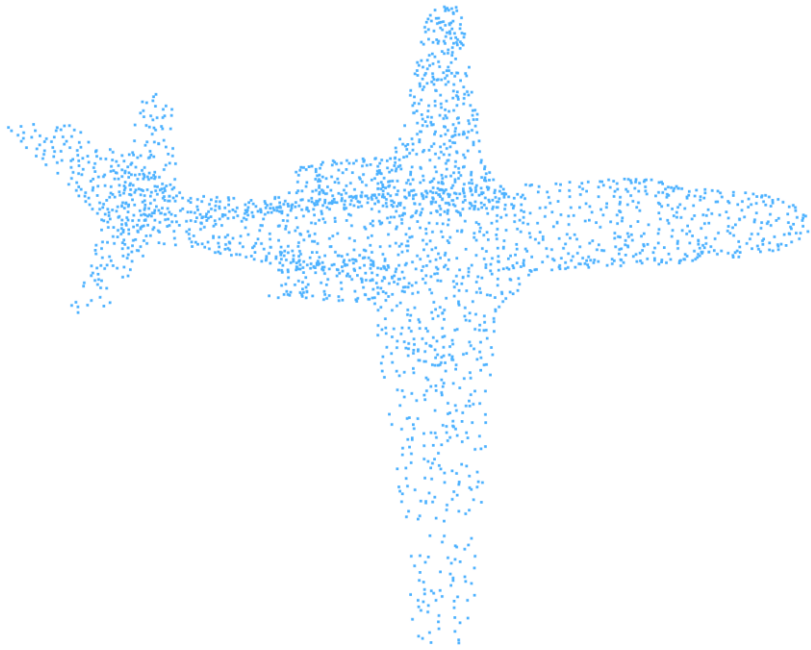


Toward per-point predictions



Segmentation Network

II. Domain Gap Reduction



Airplane from the Shapenet dataset



Airfoil from the Airfrans dataset

RQ3 Does reducing geometric distribution mismatch improve the quality of extracted features?

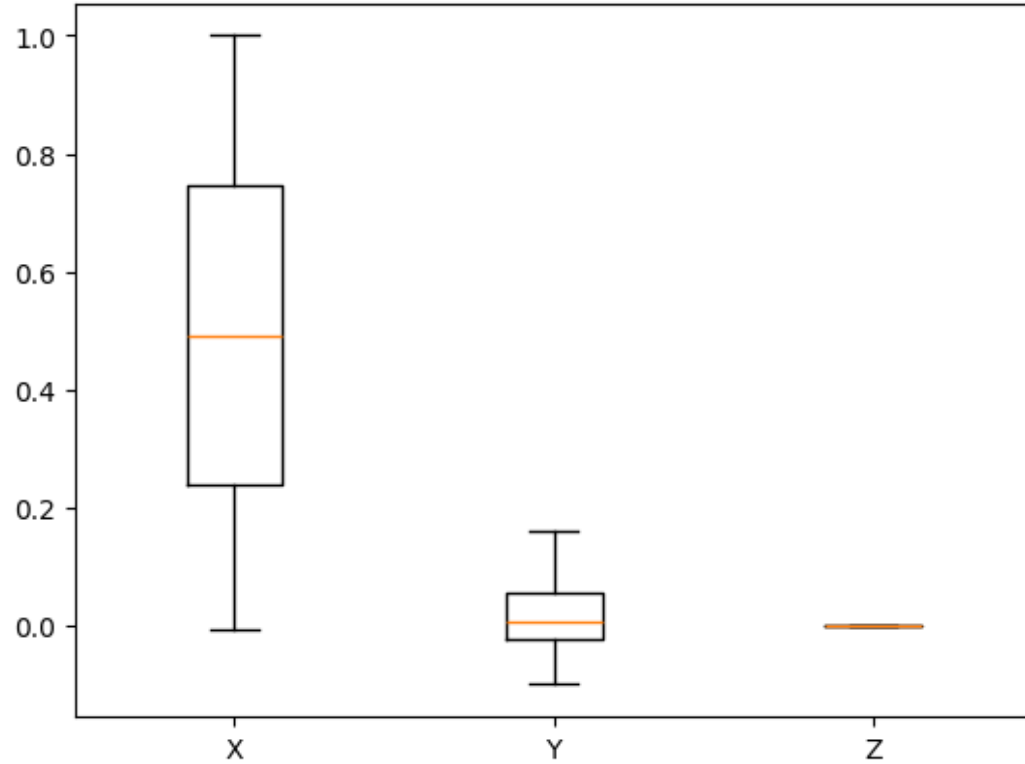
Methods:

- Shapenet dataset / Airfrans dataset comparison
- Domain gap reduction via:
 - 2D transformations: centering, standardizing
 - 3D transformations: extrusion, torus embedding

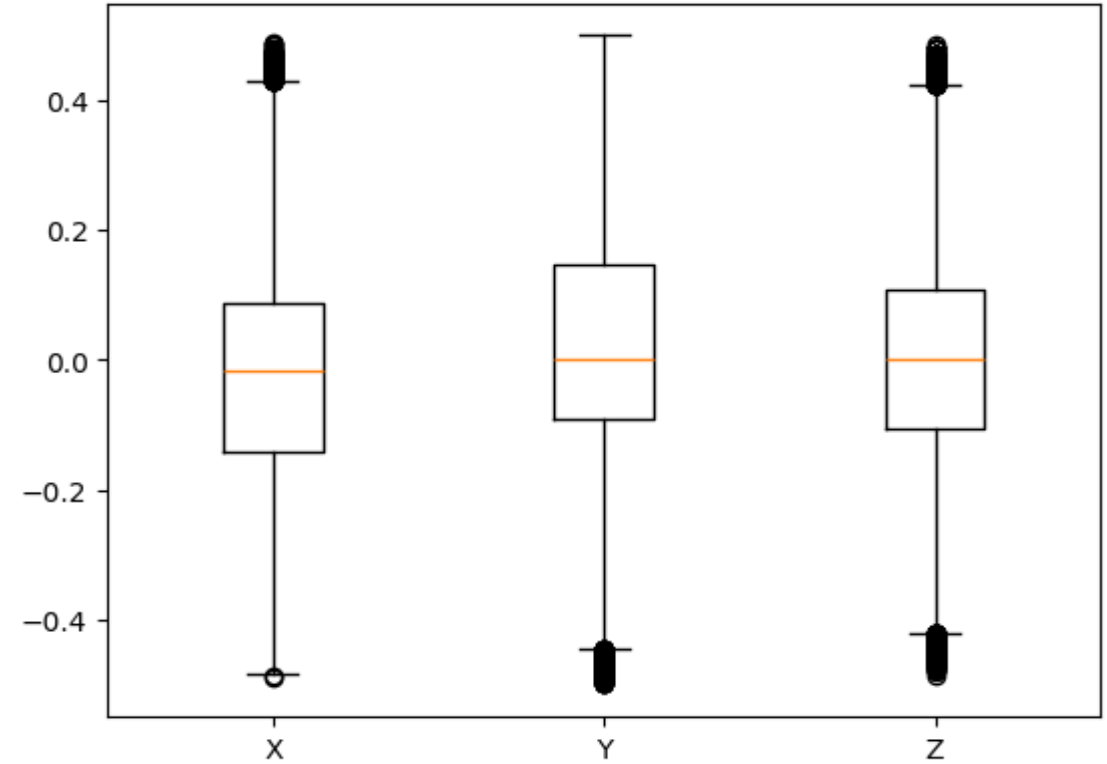
Expected results:

- Geometric transformations are expected to reduce test loss

Global X / Y / Z Boxplot for the Airfrans Dataset



Global X / Y / Z Boxplot for the Shapenet Dataset

**Observations:**

- Anisotropy VS Isotropy
- 2D VS 3D
- x centered on 0.5 VS x centered on 0

2D transformations:

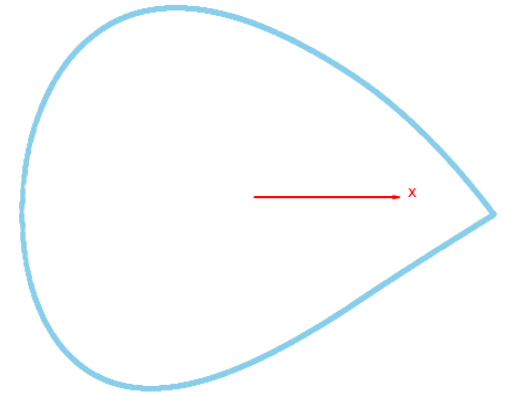
Default



Centered

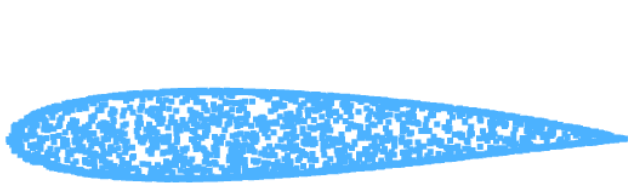


Standardized

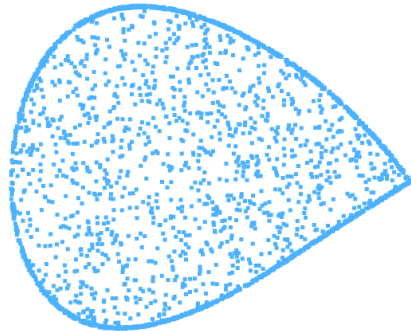


3D transformations:

Default Extrusion:



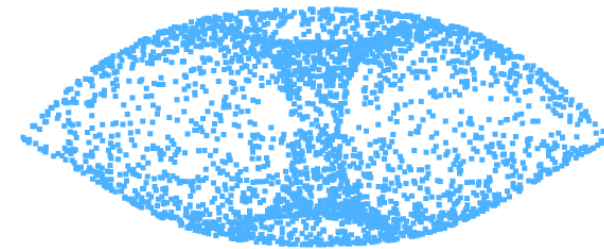
Standardized Extrusion:



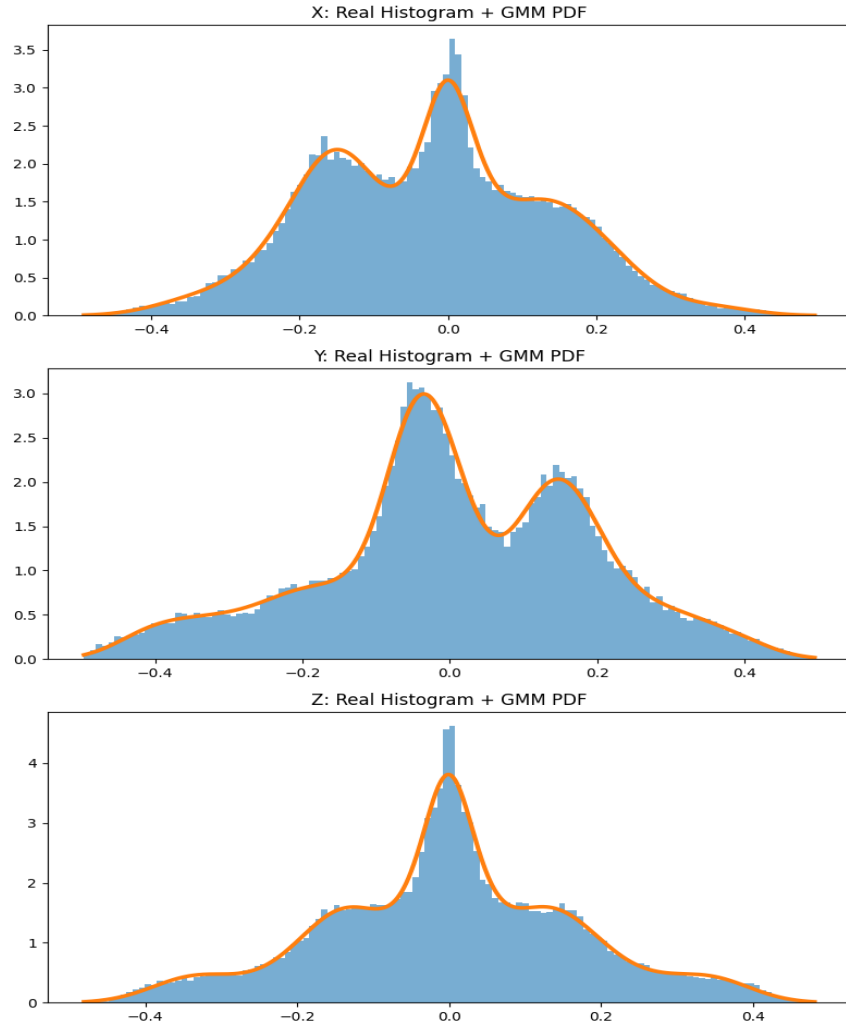
Default Torus:



Standardized Torus:



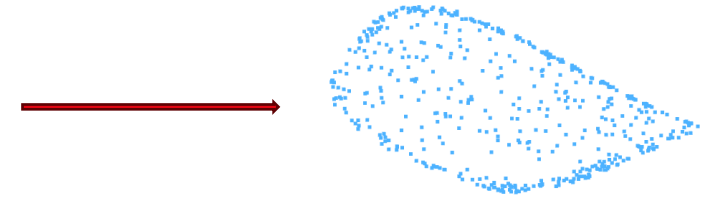
1/ Use GMM to reproduce Shapenet's distribution



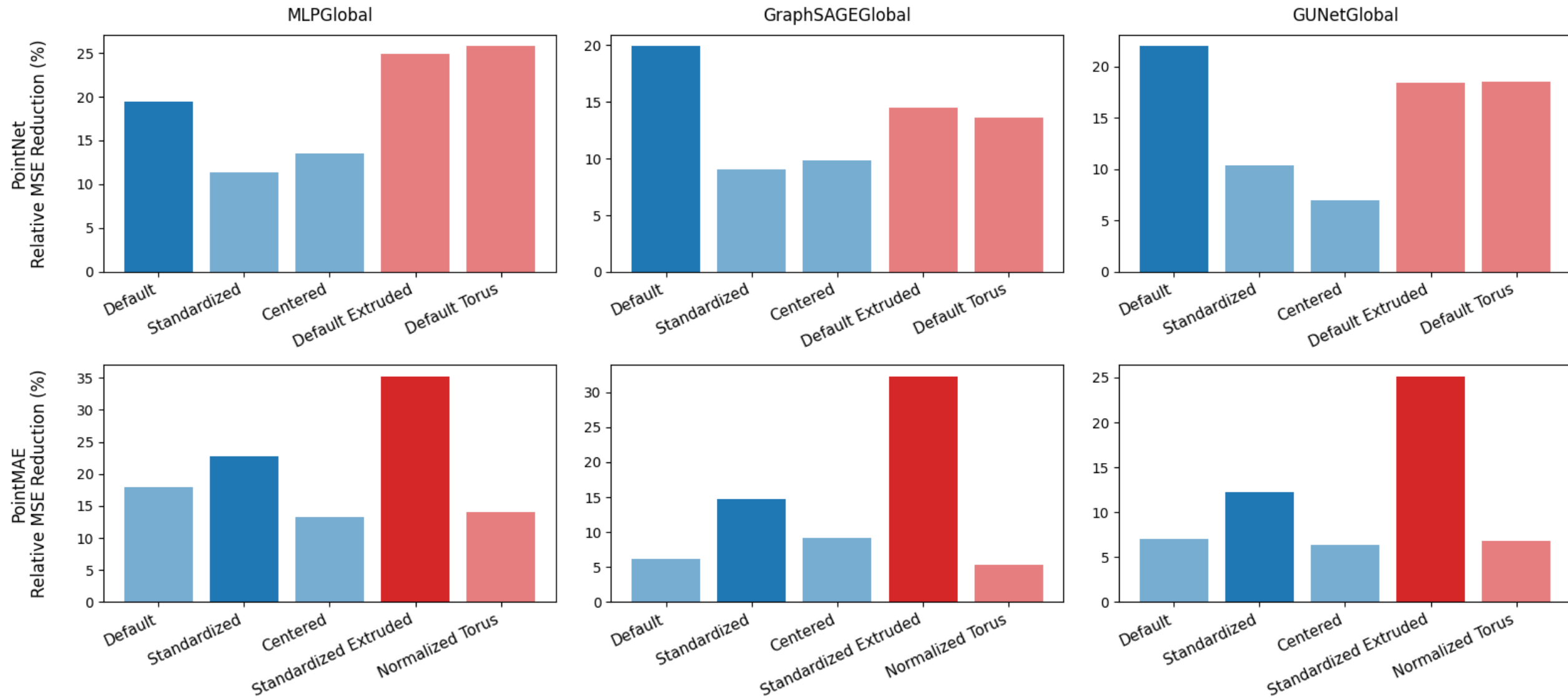
2/ Build a reference cloud with the right distribution



3/ Apply OT on extruded default airfoils to obtain deformed airfoils

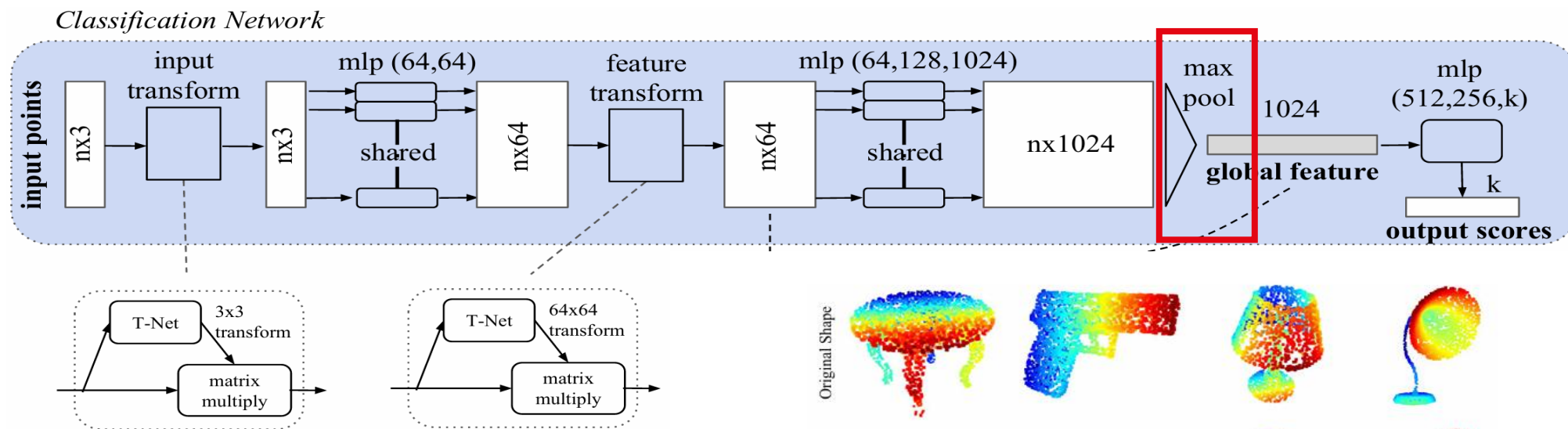


Test MSE gain for the 3 surrogate models with different extracted features

2D
3D

Why Pointnet's features are not improved by airfoil's transformation, while PointMAE's features are?

PointNet's architecture:



PointNet:

- is pretrained for classification
- relies on shared MLPs + max pooling

→ Transformations do not modify or can even dilute the information exploited by PointNet

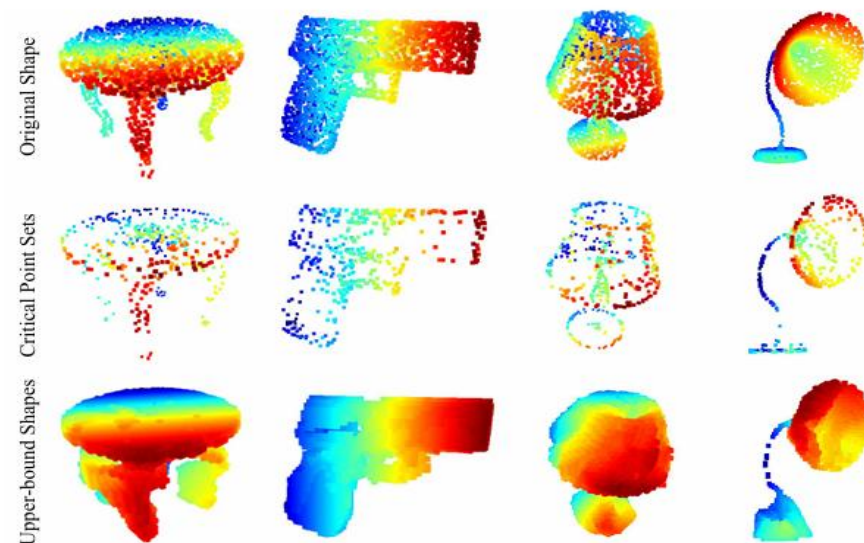


Figure 7. **Critical points and upper bound shape.** While critical points jointly determine the global shape feature for a given shape, any point cloud that falls between the critical points set and the upper bound shape gives exactly the same feature. We color-code all figures to show the depth information.

Why Pointnet's features are not improved by airfoil's transformation, while PointMAE's features are?

PointMAE:

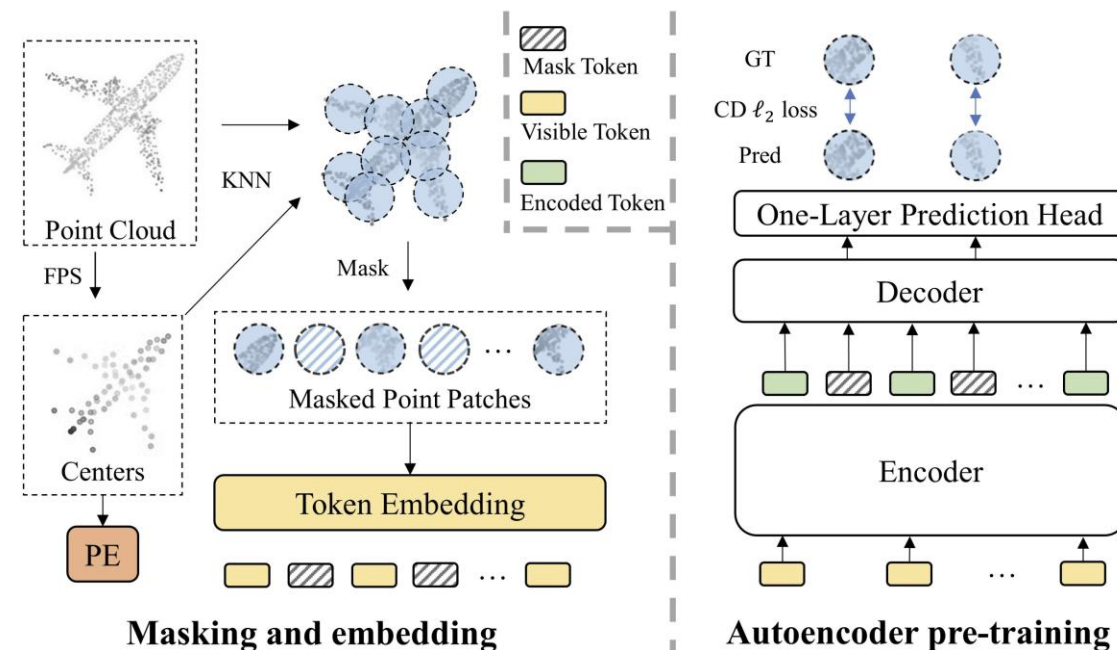
- is pretrained for reconstruction
- relies on Patch-based masked autoencoder with self-attention

→ Learns spatial relationships and global geometry

Geometric transformations:

- Amplifies geometric differences between airfoils
- Align coordinate statistics with pretraining distribution
- Introduce volumetric structure

PointMAE's architecture:



Key Takeaways:

- Pretrained 3D models learn transferable geometric representations despite domain mismatch
- Global geometric features can improve surrogate models
- Benefit of geometric alignment depends on the 3D foundation model's architecture and its pretraining task

Limitations:

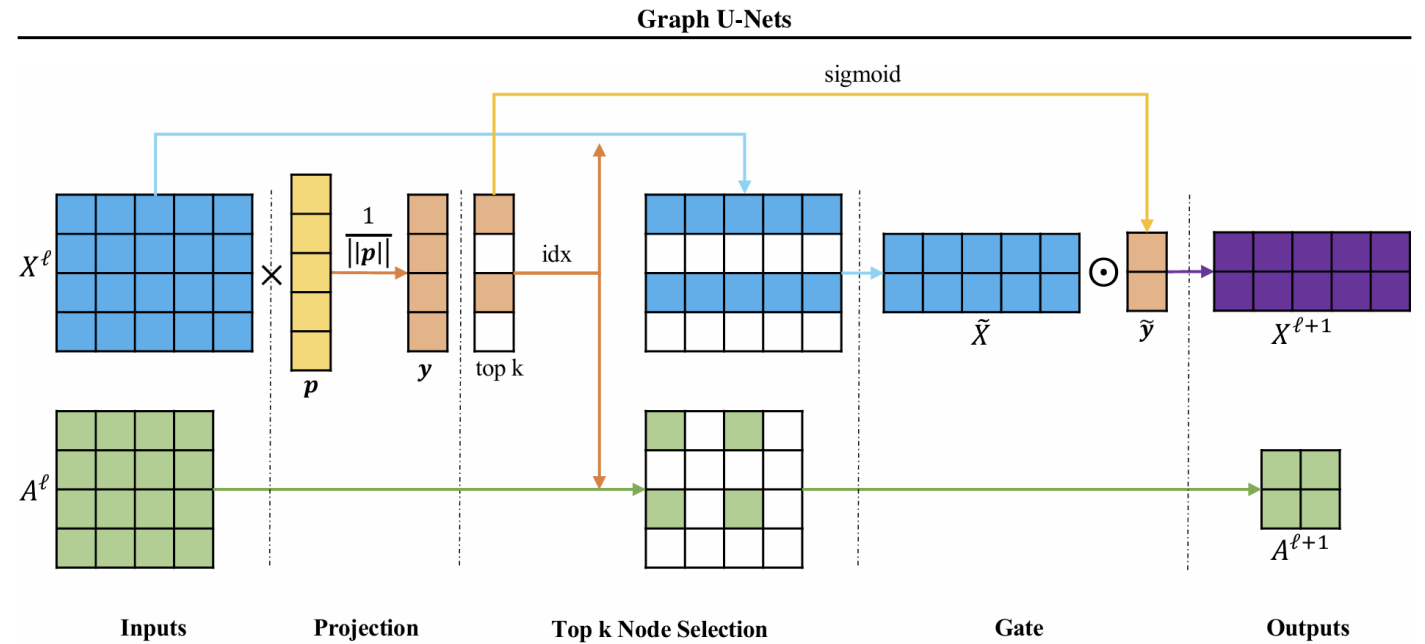
- Surrogate performance gain was used as indirect proxy for feature quality
- Global features fused via simple mechanisms, likely underestimating their full potential

Appendices

Baselines:

Graph U-Nets

Graph Pooling Layer:



Graph Unpooling Layer:

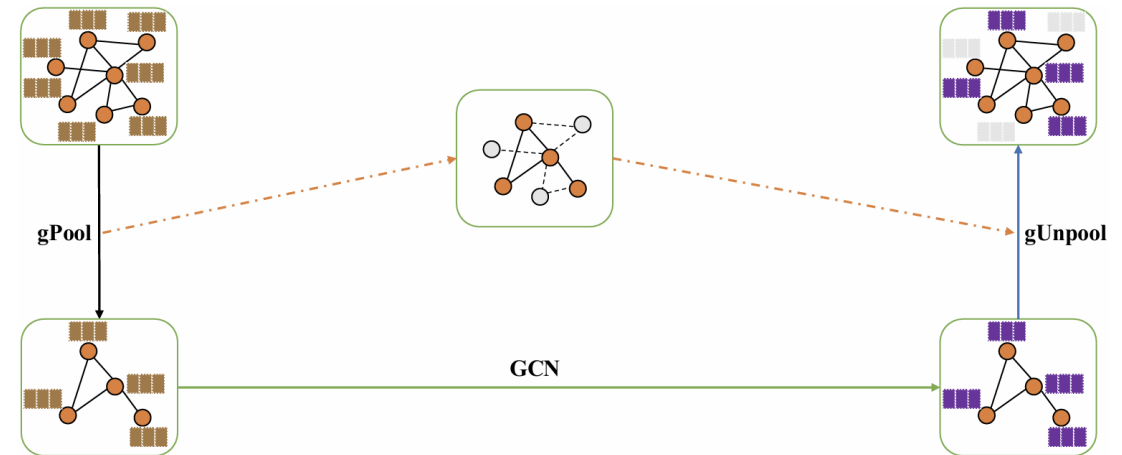


Figure 2. An illustration of the proposed graph unpooling (gUnpool) layer. In this example, a graph with 7 nodes is down-sampled using a gPool layer, resulting in a coarsened graph with 4 nodes and position information of selected nodes. The corresponding gUnpool layer uses the position information to reconstruct the original graph structure by using empty feature vectors for unselected nodes.

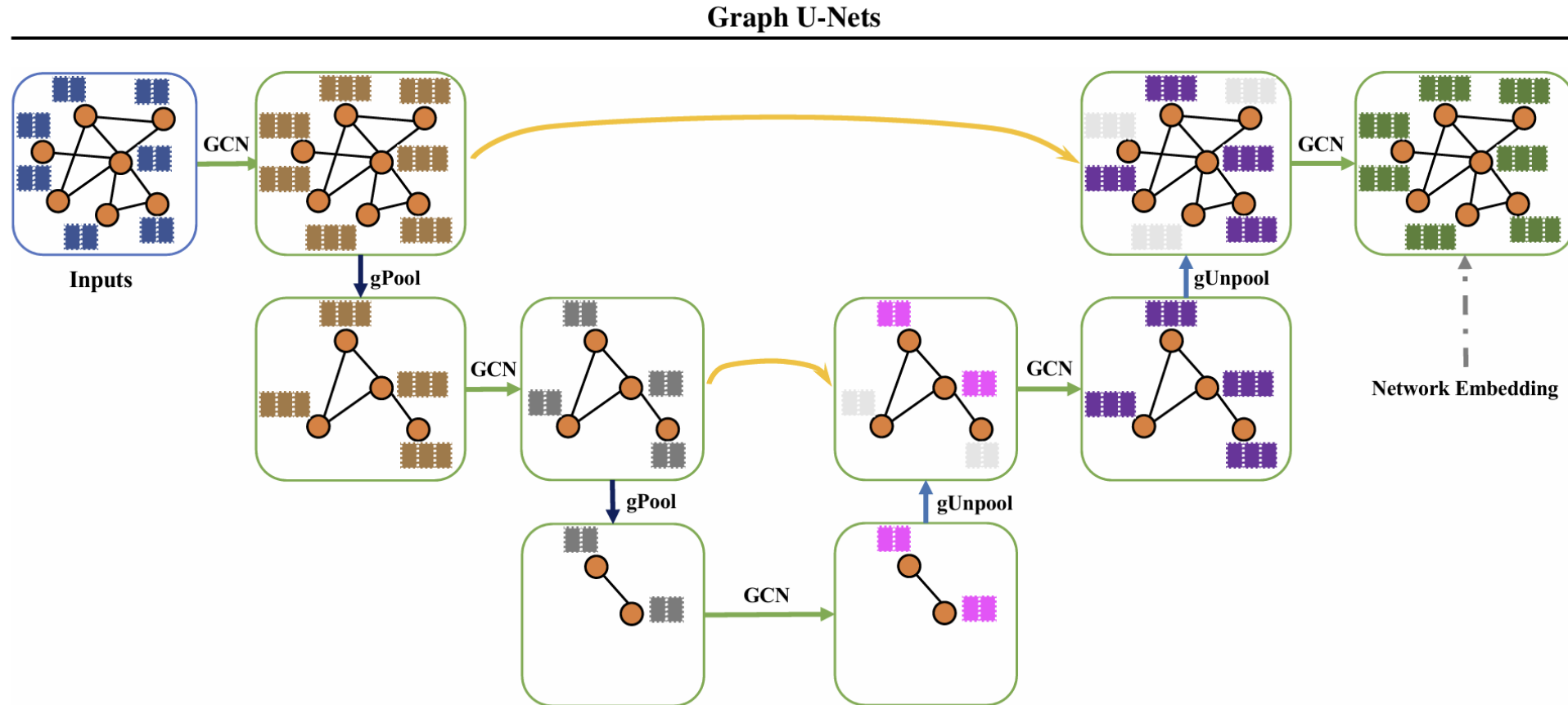


Figure 3. An illustration of the proposed graph U-Nets (g-U-Nets). In this example, each node in the input graph has two features. The input feature vectors are transformed into low-dimensional representations using a GCN layer. After that, we stack two encoder blocks, each of which contains a gPool layer and a GCN layer. In the decoder part, there are also two decoder blocks. Each block consists of a gUnpool layer and a GCN layer. For blocks in the same level, encoder block uses skip connection to fuse the low-level spatial features from the encoder block. The output feature vectors of nodes in the last layer are network embedding, which can be used for various tasks such as node classification and link prediction.

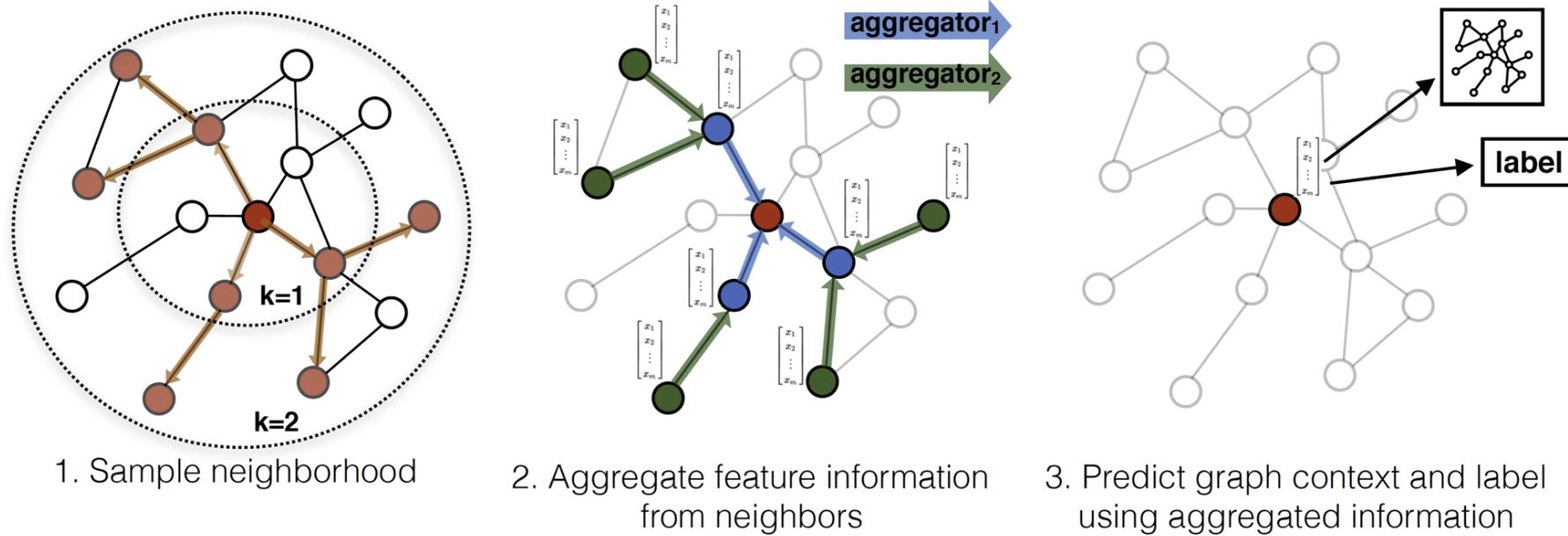


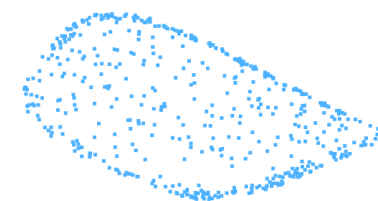
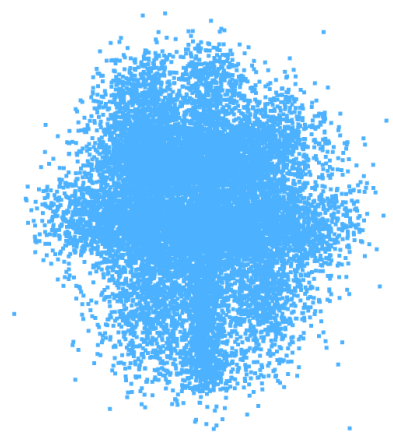
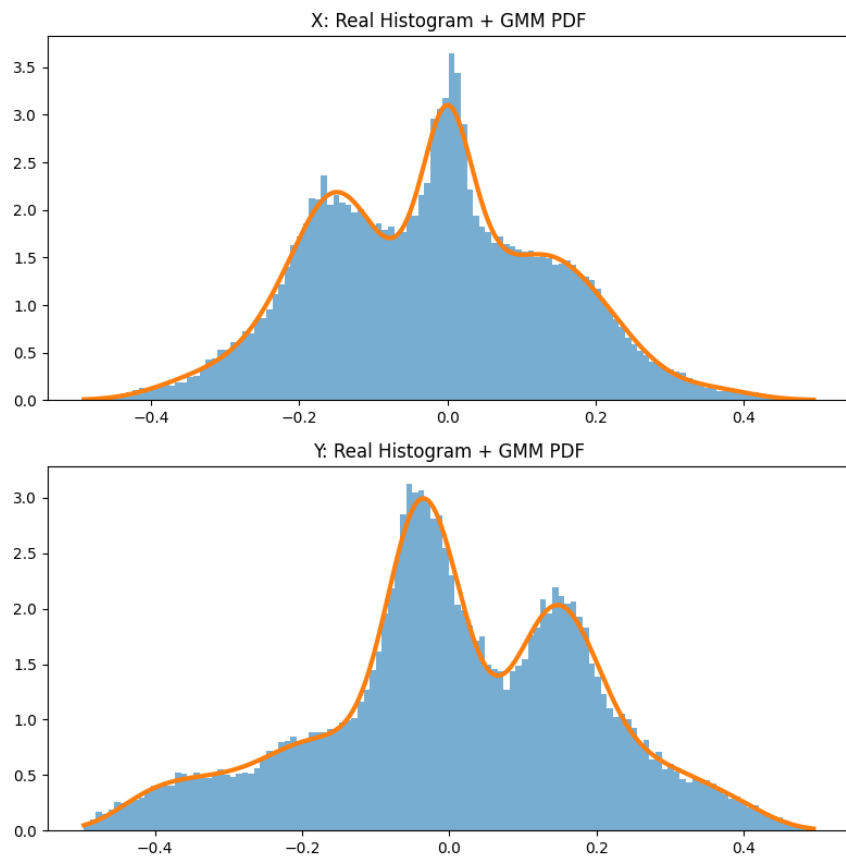
Figure 1: Visual illustration of the GraphSAGE sample and aggregate approach.

We also tried to learn a transformation:

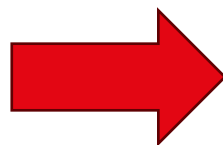
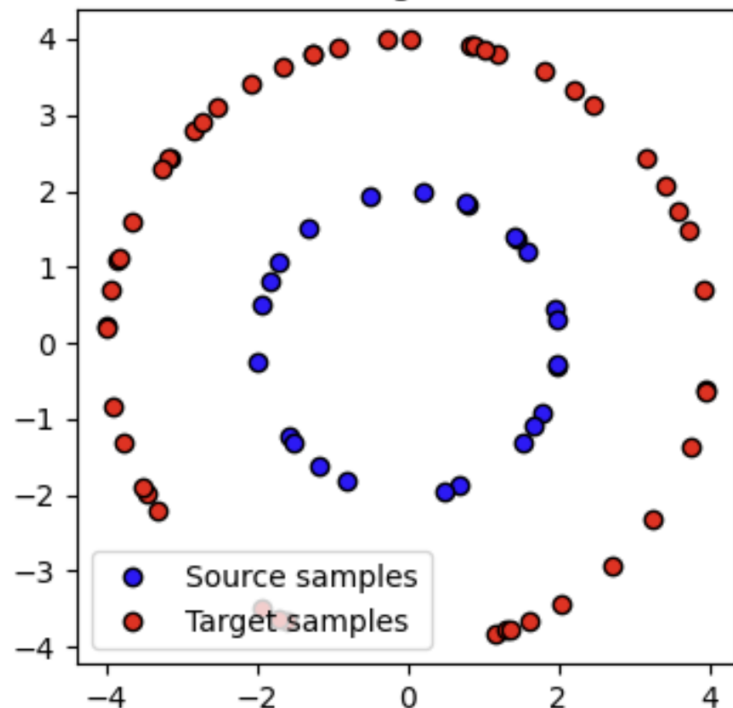
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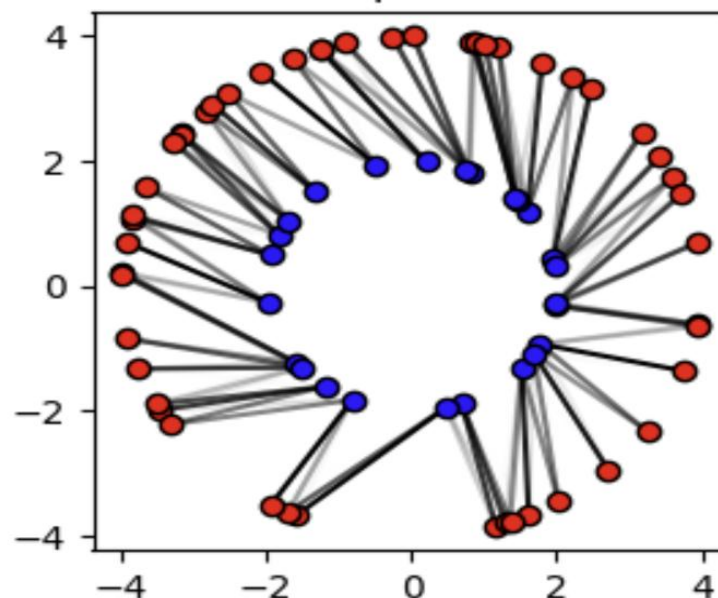
3/ Apply OT on extruded default airfoils to obtain deformed airfoils



Source and target distributions

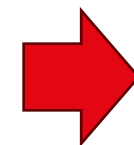


Sinkhorn OT plan loss=5.566

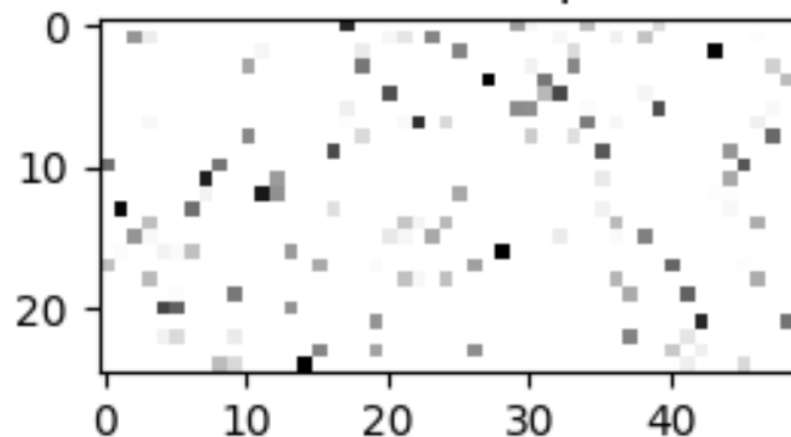


For each point i ,
Barycentric projection:

$$(X_{ot})_i = \frac{\sum_j P_{ij} Y_j}{\sum_j P_{ij}}$$



Sinkhorn OT plan



Cost matrix C

